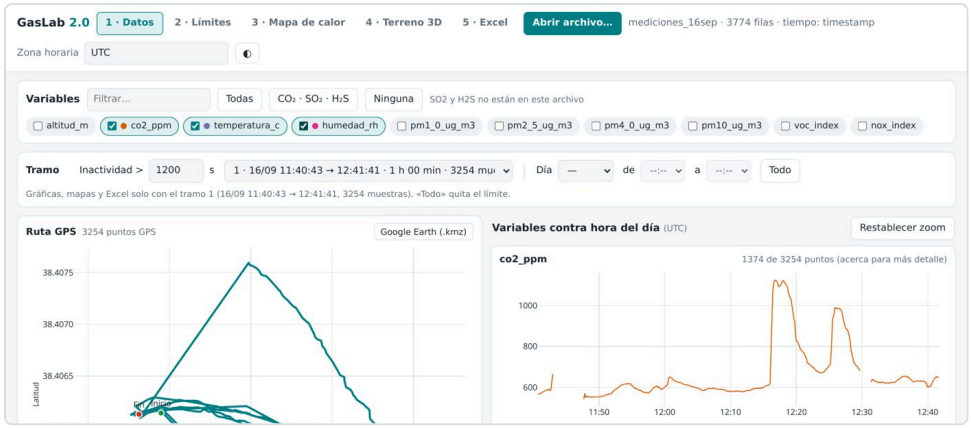
DATA → LIMITS → HEAT MAP → 3D TERRAIN → EXCEL

GasLab 2.0 · Quick guide

Open a measurement file, check the variables against time and the GPS route, and export heat maps, 3D views and Excel. The original file is never modified.

1 Data

VARIABLES, SEGMENT, ROUTE AND CHARTS



- **Variables:** tick the ones you want to plot. `CO₂ · SO₂ · H₂S` ticks the gases if the file has them.
- **Tramo** *(segment)*: a pause longer than **Inactividad > 1200 s** splits the measurement into segments. You can also pick the day and the hours. **Todo** removes the limit.
- The chosen segment applies to **all tabs**: charts, maps and Excel.
- **Drag** pans the chart and the **mouse wheel** zooms in or out. The other charts follow.
- **Double-click** or **Restablecer zoom** returns to the full view.
- Hovering marks the same instant on every chart and on the route.
- Export the route with **Google Earth (.kmz)**.

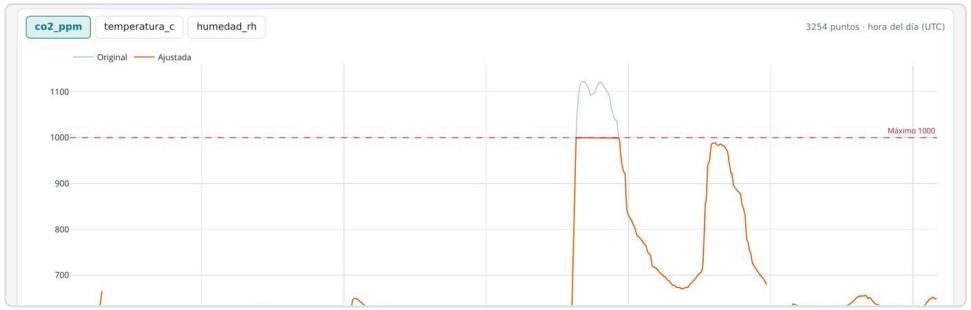
Getting started

OPEN AND SET UP

- Start with `GasLab.exe` on Windows or `./iniciar.sh` on Linux. It opens in the browser.
- Click **Abrir archivo...** *(Open file)* or drag the file onto the page. Accepts CSV (MicroGas, DJI, Campbell `Flight_Data`), `.dat` TOA5 and Excel `.xlsx`.
- At the top choose the **Zona horaria** *(time zone)* (UTC by default, or e.g. `America/Costa_Rica`) and the light or dark theme ◐.
- If the file includes them, **Original / Calibrado** and **GPS sensor / Dron / Ambos** also appear.

2 Limits

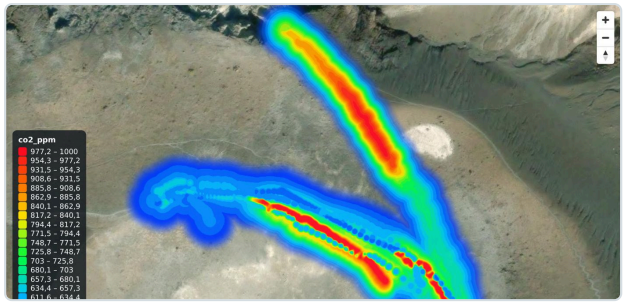
SATURATION PER VARIABLE

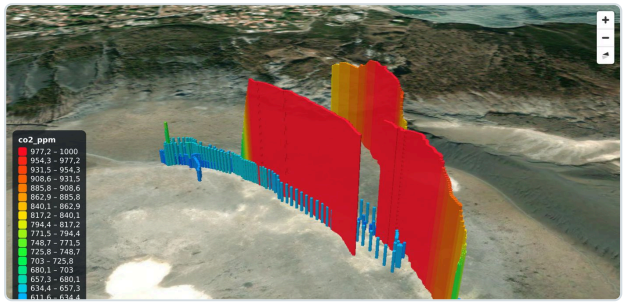


- Pick the **Dato** *(variable)* and enter a **Máximo** or **Mínimo**. Values outside the range are clamped to the limit. Leave the field empty for no limit.
- No sample is discarded. **Comparar con original** shows both curves, and the next steps use the adjusted values.

3 Heat map

2D CLOUD OR 3D BARS OVER SATELLITE



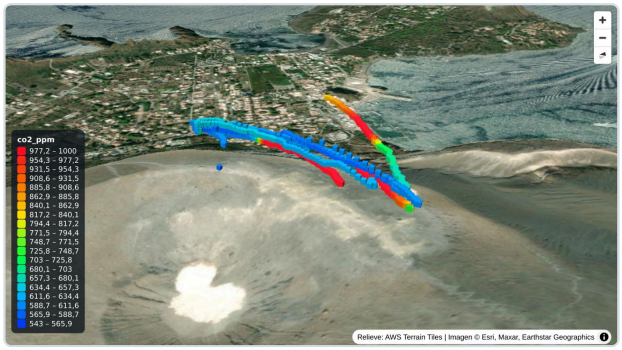


- Pick the **Señal** *(signal)* and the view `2D` or `3D`.
- **Escala** *(scale)*: linear or exponential, and how many color steps it uses.
- In **Nube** *(cloud)* you set size, opacity, cooling and reach. Each `i` button explains its option.

SAVE `Imagen PNG` `GeoTIFF + QGIS (.zip)` `Google Earth (.kmz)`

4 3D terrain

SAMPLES OVER THE REAL RELIEF



- **Altura** *(height)*: `Medida` uses the GPS altitude. `Pegada al terreno` places the samples on the ground.
- **Desfase** *(offset)* raises or lowers the samples, in meters.
- **Right-click** or **Ctrl + drag** rotates and tilts the view.

If the points end up buried or floating in Google Earth, fix it with the offset before exporting the `.kmz`.

5 Excel

EXPORT THE DATA AND THE SUMMARY



- **Tramo** *(segment)*: `Todo el archivo` (whole file), `Tramo elegido` (chosen segment) or `Parte visible (paso 1)`, which exports what is visible at the step 1 zoom.
- The workbook contains the variables ticked in step 1, **already adjusted with the step 2 limits**, using the chosen time zone and GPS.
- You can preview it first in the **Resumen**, **Datos**, **Gráficas** and **Ruta** tabs.
- Click `Descargar Excel` to save the `.xlsx`.

Internet: only for map backgrounds and relief. Your data is processed on your computer.

Old Excel: save `.xls` files as `.xlsx` or `.csv` before opening them.

Excel with several sheets: choose which one to open in the **Hoja:** *(sheet)* selector next to the file name.

Button labels are shown as they appear in the app (Spanish), with an English gloss in *italics*.